Chapter 8 : Inference for propotions

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⇒ When we collect data on categorical variables, our data consists of counts & percentages.

⇒ The parameters we want to do inference about in these setting are called population proportions.

⇒ Inference about one or two proportions is very similar to inference about means.

⇒ In particular, inference for both means & proportions is based on sampling distributions that are approximately Normal.

° Inference for a single proportion :-

CASE 8.1 **Adults and Video Games** A PriceWaterhouseCooper report estimates that the U.S. video game market was approximately $8.6 billion in 2007 and is expected to increase at an annual rate of 6.3% through 2012.[1] Who plays video games? A PEW survey, conducted by Princeton Survey Research International, reports that over half of American adults aged 18 and over play video games.[2] The PEW survey used a nationally representative sample of 2054 adults. Of the total, 1063 adults said that they played video games.

⇒ For a problems involving a single proportion, we will use n for sample size & X for the count of the outcome of interest.

↳ Often we'll use the terms "success" & "failure" for the two possible outcomes, here X is the number of successes.

EXAMPLE 8.1 Data for the Video Game Case

CASE 8.1 The *count* of people who responded "Yes" to the question about whether or not they played video games in the sample of Case 8.1 is ~~$X = 1063$. The sample size is $n = 2054$.~~

⇒ We would like to know the proportion of video games players in the adult U.S population.

↳ This population proportion is the parameter of interest.

↳ The statistics used to estimate this parameter is the sample proportion.

↳ The sample proportion is $\hat{p} = X/n$.

EXAMPLE 8.2 Estimating the Proportion of Adults Who Play Video Games

CASE 8.1 The sample proportion $\hat{p}$ in Case 8.1 is a discrete random variable that can take the values 0, 1/2054, 2/2054, . . . , 2053/2054, or 1. For our particular sample, we have

$$\hat{p} = \frac{1063}{2054} = 0.52$$

⇒ In many cases, a probability model for $\hat{p}$ can be based on the binomial distributions for counts when n is sufficiently small, we can use tables and confidence intervals for p on the discrete distribution of $\hat{p}$.

↳ We'll focus on the situation where the sample size is sufficiently large that we can approximate the distribution of $\hat{p}$ by a Normal Distribution.

Sampling Distribution of a Sample Proportion

Choose an SRS of size n from a large population that contains population proportion p of "successes." Let X be the count of successes in the sample and let $\hat{p}$ be the **sample proportion** of successes,

$$\hat{p} = \frac{X}{n}$$

Then:

- For large sample sizes, the distribution of $\hat{p}$ is **approximately Normal.**
- The **mean** of the distribution of $\hat{p}$ is p.
- The **standard deviation** of $\hat{p}$ is

$$\sqrt{\frac{p(1-p)}{n}}$$

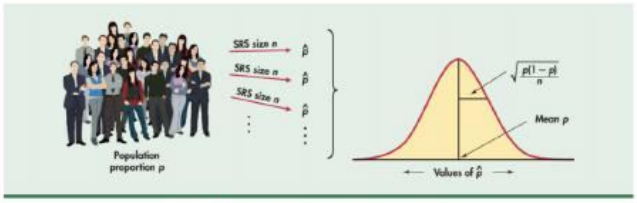


FIGURE 8.1 Draw a large SRS from a population in which the proportion p are successes. The sampling distribution of the sample proportion $\hat{p}$ of successes has approximately a Normal distribution.

- Large-sample confidence interval for a single proportion.

⇒ The sample proportion $\hat{p} = \frac{X}{n}$ is the natural estimator of the population proportion p.

↳ Because $\hat{p}$ is the mean of the sampling distribution.

⇒ The standard deviation $\sqrt{p(1-p)/n}$, the standard deviation of $\hat{p}$, depends upon the unknown parameter p.

↳ In our calculation we replace p with $\hat{p}$.

∴ Our standard error is $SE_{\hat{p}} = \sqrt{\hat{p}(1-\hat{p})/n}$

⇒ If the sample size is large, the distribution of $\hat{p}$ will be approximately Normal with mean p & standard deviation $SE_{\hat{p}}$.

z Confidence Interval for a Population Proportion

Choose an SRS of size n from a large population with unknown proportion p of successes. The **sample proportion** is

$$\hat{p} = \frac{X}{n}$$

The **standard error of $\hat{p}$** is

$$SE_{\hat{p}} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

and the **margin of error** for confidence level C is

$$m = z^* SE_{\hat{p}}$$

where z^* is the value for the standard Normal density curve with area C between $-z^*$ and z^*. The **large-sample level C confidence interval** for p is

$$\hat{p} \pm m$$

You can use this interval for 90% ($z^* = 1.645$), 95% ($z^* = 1.960$), or 99% ($z^* = 2.576$) confidence when the number of successes and the number of failures are both at least 15.

EXAMPLE 8.3 Confidence Interval for the Proportion of Adults Who Play Video Games

CASE 8.1 The sample survey in Case 8.1 found that 1063 of a sample of 2054 adults reported that they

played video games. So, the sample size is $n = 2054$ and the count is $X = 1063$. The sample proportion of adults who play video games is

$$\hat{p} = \frac{X}{n} = \frac{1063}{2054} = 0.51753$$

The standard error is

$$SE_{\hat{p}} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = \sqrt{\frac{0.5175(1-0.5175)}{2054}} = 0.011026$$

The z critical value for 95% confidence is $z^* = 1.96$, so the margin of error is

$$m = 1.96SE_{\hat{p}} = (1.96)(0.011026) = 0.021610$$

The confidence interval is

$$\hat{p} \pm m = 0.52 \pm 0.02$$

We are 95% confident that between 50% and 54% of adults play video games.

- Plus four confidence interval for a single proportion :-

⇒ Suppose we have a sample where the count is $X = 0$.

→ Then because $\hat{p} = 0$, the SE and margin of error (ME) would be a single point 0.

→ Confidence interval based on the large-sample Normal approximation do not make sense in this situation.

⇒ We can do better by moving the sample proportion $\hat{p}$ slightly away from 0 and 1.

→ We can do it in several ways.

→ Idea: Act as if we have 4 additional observations, 2 of which are successes & 2 of which are failure.

→ The new sample size size is $n+4$ & the count of success is $X+2$.

→ This is called Wilson estimate.

To compute a confidence interval based on the Wilson estimate, first replace the value of X by $X + 2$ and the value of n by $n + 4$. Then use these values in the formulas for the z confidence interval.

In Example 8.1, we had $X = 1063$ and $n = 2054$. To apply the plus four approach we use the z procedure with $X = 1065$ and $n = 2058$. You can use this interval when the sample size is at least $n = 10$ and the confidence level is 90%, 95%, or 99%.

- Significance test for a single proportion :

z Significance Test for a Population Proportion

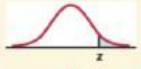
Choose an SRS of size n from a large population with unknown proportion p of successes. To test the hypothesis H_0: $p = p_0$, compute the z **statistic**

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

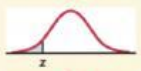
p_0 not $\hat{p}$!

In terms of a standard Normal random variable Z, the approximate P-value for a test of H_0 against

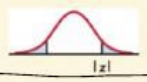
H_a: $p > p_0$ is $P(Z \geq z)$



H_a: $p < p_0$ is $P(Z \leq z)$



H_a: $p \neq p_0$ is $2P(Z \geq |z|)$



Use this test when the expected number of successes np_0 and the expected number of failures $n(1 - p_0)$ are both at least 10.

EXAMPLE 8.4 Comparing Two Sun Block Lotions

Your company produces a sun block lotion designed to protect the skin from both UVA and UVB exposure to the sun. You hire a company to compare your product with the product sold by your major competitor. The testing company exposes skin on the backs of a sample of 20 people to UVA and UVB rays and measures the protection provided by each product. For 13 of the subjects, your product provided better protection, while for the other 7 subjects, your competitor's product provided better protection. Do you have evidence to support a commercial claiming that your product provides superior UVA and UVB protection? For the data we have $n = 20$ subjects and $X = 13$ successes. To answer the claim question, we test

$$H_0: p = 0.5$$
$$H_a: p \neq 0.5$$

The expected numbers of successes (your product provides better protection) and failures (your competitor's product provides better protection) are $20 \times 0.5 = 10$ and $20 \times 0.5 = 10$. Both are at least 10, so we can use the z test. The sample proportion is

$$\hat{p} = \frac{X}{n} = \frac{13}{20} = 0.65$$

The test statistic is

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} = \frac{0.65 - 0.5}{\sqrt{\frac{(0.5)(0.5)}{20}}} = 1.34$$

From Table A we find $P(Z \geq 1.34) = 0.9099$, so the probability in the upper tail is $1 - 0.9099 = 0.0901$. The P-value is the area in both tails, $P = 2 \times 0.0901 = 0.1802$. Minitab and SAS outputs for the analysis appear in Figure 8.3. We conclude that the sun block testing data are compatible with the hypothesis of no difference between your product and your competitor's ($\hat{p} = 0.65$, $z = 1.34$, $P = 0.18$). The data do not provide you with a basis to support your advertising claim.

- In settings like 8.4, we must start with the view that either product could be better if we want to prove a claim of superiority.
  - Thinking or hoping that your product is superior cannot be used to justify a one-sided test.

- Choosing a sample size:

⇒ Recall that the margin of error for the large-sample confidence interval for a population proportion is

$$m = z^* SE_{\hat{p}} = z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

⇒ Choosing a confidence level C fixes the z^* critical value.

⇒ The margin of error also depends on $\hat{p}$ & n.
  - Because we don't know the value of $\hat{p}$ until we gather the data, we must guess the value to use in calculations, will call the guessed value p^*.
    - Here's two ways to get p^*.
      1) Use sample estimates from pilot study, or from similar studies done earlier.
      2) Use $p^* = 0.5$. Because the margin of error is the largest when $\hat{p} = 0.5$, this choice gives a sample size that is larger that we really need (upper bound) for the confidence level we choose. It's a safe choice.

Sample Size for Desired Margin of Error

The level C confidence interval for a proportion p will have a margin of error approximately equal to a specified value m when the sample size is

$$n = \left(\frac{z^*}{m}\right)^2 p^*(1 - p^*)$$

Here z^* is the critical value for confidence C, and p^* is a guessed value for the proportion of successes in the future sample.

The margin of error will be less than or equal to m if p^* is chosen to be 0.5. The sample size required is then given by

$$n = \left(\frac{z^*}{2m}\right)^2$$

⇒ If your sample turns out to have $\hat{p} < 0.3$ or $\hat{p} > 0.7$, the sample size based on $p^* = 0.5$ may be much larger than needed.

EXAMPLE 8.5 Planning a Sample of Customers

Your company has received complaints about its customer support service. You intend to hire a consulting company to carry out a sample survey of customers. Before contacting the consultant, you want some idea of the sample size you will have to pay for. One critical question is the degree of satisfaction with your customer service, measured on a five-point scale. You want to estimate the proportion p of your customers who are satisfied (that is, who choose either "satisfied" or "very satisfied," the two highest levels on the five-point scale).

You want to estimate p with 95% confidence and a margin of error less than or equal to 3%, or 0.03. For planning purposes, you are willing to use $p^* = 0.5$. To find the sample size required,

$$n = \left(\frac{z^*}{2m}\right)^2 = \left[\frac{1.96}{(2)(0.03)}\right]^2 = 1067.1$$

Round up to get $n = 1068$. (Always round up. Rounding down would give a margin of error slightly greater than 0.03.)

Similarly, for a 2.5% margin of error we have (after rounding up)

$$n = \left[\frac{1.96}{(2)(0.025)}\right]^2 = 1537$$

and for a 2% margin of error,

$$n = \left[\frac{1.96}{(2)(0.02)}\right]^2 = 2401$$

CASE 8.2 **Marketing Christmas Trees** An association of Christmas tree growers in Indiana sponsored a sample survey of Indiana households to help improve the marketing of Christmas trees.[5] The researchers decided to use a telephone survey and estimated that each telephone interview would take about 2 minutes. Nine trained students in agribusiness marketing were to make the phone calls between 1:00 P.M. and 8:00 P.M. on a Sunday. After discussing

problems related to people not being at home or being unwilling to answer the questions, the survey team proposed a sample size of 500. Several of the questions asked demographic information about the household. The key questions of interest had responses of "Yes" or "No," for example, "Did you have a Christmas tree last year?" The primary purpose of the survey was to estimate various sample proportions for Indiana households. An important issue in designing the survey was therefore whether the proposed sample size of $n = 500$ would be adequate to provide the sponsors of the survey with the information they required.

To address this question, we calculate the margins of error of 95% confidence

EXAMPLE 8.6 Margins of Error

CASE 8.2 In the Christmas tree market survey, the margin of error of a 95% confidence interval for any value of $\hat{p}$ and $n = 500$ is

$$m = z^* \mathrm{SE}_{\hat{p}} = 1.96\sqrt{\frac{\hat{p}(1-\hat{p})}{500}}$$

The results for various values of $\hat{p}$ are

$\hat{p}$	m	$\hat{p}$	m
0.05	0.019	0.60	0.043
0.10	0.026	0.70	0.040
0.20	0.035	0.80	0.035
0.30	0.040	0.90	0.026
0.40	0.043	0.95	0.019
0.50	0.044		

The survey team judged these margins of error to be acceptable, and they used a sample size of 500 in their survey.

⇒ The above table in example 8.6 highlights two points.

1) The margins of error for $\hat{p} = 0.05$ & $\hat{p} = 0.95$ are same.

↳ This is direct consequence of the form of confidence interval.

2) The margin of error varies only between 0.040 & 0.044 as $\hat{p}$ varies from 0.3 to 0.7, and the error is greatest at 0.5, as was claimed earlier.

- Comparing two Proportions:

⇒ We call the two groups being compared Population-1 & Population-2, and the two proportions of "successes" p_1 & p_2.

↳ The data consists of two independent SRSs. The sample sizes are n_1 for population 1 & n_2 for population 2.

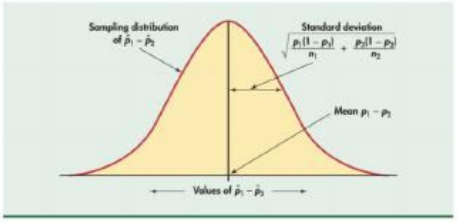
⇒ The proportion of successes in each sample estimates the corresponding population proportions.

Population	Population proportion	Sample size	Count of successes	Sample proportion
1	p_1	n_1	X_1	$\hat{p}_1 = X_1/n_1$
2	p_2	n_2	X_2	$\hat{p}_2 = X_2/n_2$

⇒ To compare the unknown population proportions, start with the observed difference between two sample proportions.

$$D = \hat{p}_1 - \hat{p}_2.$$

FIGURE 8.4 The sampling distribution of the difference between two sample proportions is approximately Normal. The mean and standard deviation are found from the two population proportions of successes, p_1 and p_2.



⇒ Because the two samples are independent, the two $\hat{p}$'s are also independent.

Sampling Distribution of $\hat{p}_1 - \hat{p}_2$

Choose independent SRSs of sizes n_1 and n_2 from two populations with proportions p_1 and p_2 of successes. Let $D = \hat{p}_1 - \hat{p}_2$ be the difference between the two sample proportions of successes. Then

- As both sample sizes increase, the sampling distribution of D becomes **approximately Normal.**
- The **mean** of the sampling distribution is $p_1 - p_2$.
- The **standard deviation** of the sampling distribution is

$$\sigma_D = \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$$

- Large-sample confidence interval for a difference in proportions:-

z Confidence Interval for Comparing Two Proportions

Choose an SRS of size n_1 from a large population having proportion p_1 of successes and an independent SRS of size n_2 from another population having proportion p_2 of successes.

The large-sample estimate of the difference in proportions is

$$D = \hat{p}_1 - \hat{p}_2 = \frac{X_1}{n_1} - \frac{X_2}{n_2}$$

The **standard error of the difference** is

$$\text{SE}_D = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

and the **margin of error for confidence level C** is

$$m = z^* \mathrm{SE}_D$$

where z^* is the value for the standard Normal density curve with area C between $-z^*$ and z^*. The **large-sample level C confidence interval** for $p_1 - p_2$ is

$$(\hat{p}_1 - \hat{p}_2) \pm m$$

Use this method when the number of successes and the number of failures in each of the samples are at least 10.



CASE 8.3

"No Sweat" Garment Labels Following complaints about the working conditions in some apparel factories both in the United States and abroad, a joint government and industry commission recommended that companies that monitor and enforce proper standards be allowed to display a "No Sweat" label on their products. Does the presence of these labels influence consumer behavior?

A survey of U.S. residents aged 18 or older asked a series of questions about how likely they would be to purchase a garment under various conditions. For some conditions, it was stated that the garment had a "No Sweat" label; for others, there was no mention of such a label. On the basis of the responses, each person was classified as a "label user" or a "label nonuser."[10] About 16.5% of those surveyed were label users. One purpose of the study was to describe the demographic characteristics of users and nonusers.

Here is a summary of the data. We let X denote the number of label users.

Population	n	X	$\hat{p} = X/n$
1 (women)	296	63	0.213
2 (men)	251	27	0.108

The study in Case 8.3 suggested that there is a gender difference in the proportion of label users. Let's explore this possibility using a confidence interval.

EXAMPLE 8.7 Gender Differences in Label Use

CASE 8.3

First, we find the estimate of the difference:

$$D = \hat{p}_1 - \hat{p}_2 = \frac{X_1}{n_1} - \frac{X_2}{n_2} = 0.213 - 0.108 = 0.105$$

Next, we calculate the standard error:

$$\mathrm{SE}_D = \sqrt{\frac{0.213(1-0.213)}{63} + \frac{0.108(1-0.108)}{27}} = 0.0308$$

For 95% confidence, we use $z^* = 1.96$, so the margin of error is

$$m = z^* \mathrm{SE}_D = (1.96)(0.0308) = 0.060$$

The large-sample 95% confidence interval is

$$D \pm m = 0.105 \pm 0.060 = (0.04,\ 0.16)$$

With 95% confidence we can say that the difference in the proportions is between 0.04 and 0.16. Alternatively, we can report that the gender difference is about 10% in favor of women, with a 95% margin of error of 6%.

- Plus four confidence intervals for a difference in proportions:

⇒ As before, we add 2 successes & 2 failures to the actual data, but this time we divide them equally between the two sample.

↳ this method is recommended when both sample size is ≥ 5, and the confidence interval is 90%, 95%, or 99%.

In Example 8.7, we had $X_1 = 63$, $n_1 = 296$, $X_2 = 27$, and $n_2 = 251$. For the plus four procedure, we would use $X_1 = 64$, $n_1 = 298$, $X_2 = 28$, and $n_2 = 253$.

- Significance tests:

⇒ It's sometimes useful to test the null hypothesis that the two population proportions are same.

⇒ We standardize $D = \hat{p}_1 - \hat{p}_2$ by subtracting its mean $p_1 - p_2$ & then dividing by its standard deviation

$$\sigma_D = \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$$

⇒ If n_1 & n_2 are large, the standardized difference is approximately $N(0,1)$.

⇒ for hypothetical test, we follow the common practice of replacing the unknown $\sigma_{\bar{D}}$ with an estimate that takes in account the null hypothesis that $p_1 = p_2$.

⇒ So if $p_1 = p_2$, let p denote the common value. The standard deviation of $D = \hat{p}_1 - \hat{p}_2$ is then.

$$\sigma_{\bar{D}} = \sqrt{\frac{p(1-p)}{n_1} + \frac{p(1-p)}{n_2}}$$

$$\sigma_{\bar{D}} = \sqrt{p(1-p)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

⇒ The subscript 0p reminds of special condition where $p_1 = p_2 = p$.

⇒ We estimate the common value of p by the overall proportion of successes in the two samples:

$$\hat{p} = \frac{\text{number of successes in both samples}}{\text{number of observations in both samples}} = \frac{x_1 + x_2}{n_1 + n_2}.$$

→ This estimate is called the pooled estimate.

Significance Tests for Comparing Two Proportions

Choose an SRS of size n_1 from a large population having proportion p_1 of successes and an independent SRS of size n_2 from another population having proportion p_2 of successes. To test the hypothesis

$$H_0: p_1 = p_2$$

compute the **z statistic**

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\text{SE}_{0p}}$$

where the **pooled standard error** is

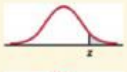
$$\text{SE}_{0p} = \sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

based on the **pooled estimate** of the common proportion of successes

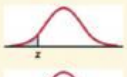
$$\hat{p} = \frac{X_1 + X_2}{n_1 + n_2}$$

In terms of a standard Normal random variable Z, the P-value for a test of H_0 against

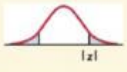
$H_a: p_1 > p_2$ is $P(Z \geq z)$



$H_a: p_1 < p_2$ is $P(Z \leq z)$



$H_a: p_1 \neq p_2$ is $2P(Z \geq |z|)$



Use this test when the number of successes and the number of failures in each of the samples are at least 5.

EXAMPLE 8.8 Men, Women, and Garment Labels

CASE 8.3 Example 8.7 (page 475) presents survey data on whether consumers are "label users" who pay attention to label details when buying a garment. Are men and women equally likely to be label users? Here is the data summary:

Population	n	X	$\hat{p} = X/n$
1 (women)	296	63	0.213
2 (men)	251	27	0.108

The sample proportions are certainly quite different, but we need a significance test to verify that the difference is too large to easily result from the play of chance in choosing the sample. Formally, we compare the proportions of label users in the two populations (women and men) by testing the hypotheses

$$H_0: p_1 = p_2$$
$$H_a: p_1 \neq p_2$$

The pooled estimate of the common value of p is

$$\hat{p} = \frac{63+27}{296+251} = \frac{90}{547} = 0.1645$$

This is just the proportion of label users in the entire sample.

First, we compute the standard error

$$\text{SE}_{Dp} = \sqrt{(0.1645)(0.8355)\left(\frac{1}{296}+\frac{1}{251}\right)} = 0.03181$$

and then we use this in the calculation of the test statistic

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\text{SE}_{Dp}} = \frac{0.213 - 0.108}{0.03181} = 3.30$$

The difference in the sample proportions is more than 3 standard deviations away from zero. The P-value is $2P(Z \geq 3.30)$. From Table A we have $P = 2 \times 0.0005 = 0.0010$. Software gives $P = 0.0009$. We report: 21% of women are label users versus only 11% of men; the difference is statistically significant ($z = 3.30$, $P < 0.001$).

- Beyond the Basics: Relative Risk:

⇒ We compared the proportions by giving confidence interval for difference of proportions.

↳ Alternatively we might choose to make this comparison by giving the ratio of the two proportions.

↳ This ratio is called relative risk (RR).

⇒ Confidence interval for RR applies the principles that we have studied, but the details are somewhat complicated.

↳ Book leaves the detail to software & concentrate on interpreting & communicating the results.

EXAMPLE 8.9 Relative Risk for Use of Labels

The following table summarizes the data on the proportions of men and women who use labels when buying clothing:

Population	n	X	$\hat{p} = X/n$
1 (women)	296	63	0.2128
2 (men)	251	27	0.1076

The relative risk for this sample is

$$\text{RR} = \frac{\hat{p}_1}{\hat{p}_2} = \frac{0.2128}{0.1076} = 1.98$$

Confidence intervals for the relative risk in the entire population of shoppers are based on this sample relative risk. Software (for example, PROC FREQ with the MEASURES option in SAS) gives a 95% confidence interval as 1.30 to 3.01. Our summary: Women are about twice as likely as men to use labels; the 95% confidence interval is (1.30, 3.01).

EXAMPLE 8.10 Smoking and Colorectal Cancer

Colorectal cancer is fourth in the list of types of cancers that lead to death. Many studies have examined the relationship between cigarette smoking and colorectal cancer but the results have been inconsistent. Twenty-six studies gave relative risk estimates for people who had ever smoked relative to those who had never smoked. A recent study combined the results of these studies to obtain a summary measure of relative risk.[20] The smokers are Population 1 and the nonsmokers are Population 2. The report of the study stated that the relative risk was 1.18 with a 95% confidence interval of 1.11 to 1.25. Since the confidence interval does not include the value of 1, which would correspond to equal risks in the two populations, we conclude that there is a higher risk of colorectal cancer for cigarette smokers. The estimated increase in risk is 18% with a 95% confidence interval of 11% to 25%.